$\Rightarrow \quad mc + mc - 5m + 6c - 30 + mc - 10m + 12c - 120 + mc - 15m + 18c - 270 + mc$
$- 20m + 24c - 480 = 545$

$\Rightarrow \quad 5mc - 50m + 60c = 1445$

$\Rightarrow \quad mc - 10m + 12c = 289$

$\Rightarrow \quad mc - 10m = 289 - 12c$

$\Rightarrow \; m = \dfrac{289 - 12c}{c - 10}$.

Clearly, $c > 20$
[$\because$ if $c < 20$, then number of crates shifted on 5th day i.e., $(c - 20) < 0$ which is not possible]
Thus, we have to find $c > 20$ such that m is an integer.
This is not so for $c = 21$ and $c = 22$.

For $c = 23$, we have: $m = \dfrac{289 - 276}{13} = \dfrac{13}{13} = 1$.

519. Number of crates shifted on third day $= (m + 12)(c - 10)$
$= (1 + 12)(23 - 10)$
$= 13 \times 13 = 169$.

520. Number of men on 5th day $= m + 24 = 25$.

521. Let the missing number be x.

Given $\dfrac{20 + 8 \times 0.5}{20 - x} = 12$

$\Rightarrow \dfrac{20+4}{20-x} = 12 \Rightarrow \dfrac{24}{20-x} = 12$

$\Rightarrow 20 - x = \dfrac{24}{12} = 2$

$\Rightarrow x = 20 - 2 = 18$

Hence, the number is 18.

522. Given $\dfrac{a}{b} + \dfrac{b}{a} = 2$

$\Rightarrow \dfrac{a^2 + b^2}{ab} = 2$

$\Rightarrow a^2 + b^2 = 2ab$

$\Rightarrow a^2 + b^2 - 2ab = 0$

$\Rightarrow (a - b)^2 = 0$

$\Rightarrow a - b = 0$

523. Given $(24.96)^2 \div (34.11 \div 20.05) + 67.96 - 89.11$

$(25)^2 \div (34 \div 20) + 68 - 89$

$\approx (25)^2 \div \left(\dfrac{34}{20}\right) + 68 - 89$

$\approx 625 \div 1.7 + 68 - 89$

$\approx 367.6 + 68 - 89$

$\approx 367 + 68 - 89 \approx 346$

524. $\dfrac{x+1}{x-1} = \dfrac{a}{b}$

By componendo and dividendo

$\dfrac{x+1+x-1}{x+1-x+1} = \dfrac{a+b}{a-b}$

$\Rightarrow \dfrac{2x}{2} = \dfrac{a+b}{a-b}$

$\Rightarrow \; x = \dfrac{a+b}{a-b}$ ....(i)

Again, $\dfrac{1-y}{1+y} = \dfrac{b}{a}$

$\Rightarrow \dfrac{1+y}{1-y} = \dfrac{a}{b}$

$\Rightarrow \dfrac{1+y+1-y}{1+y-1+y} = \dfrac{a+b}{a-b}$

$\Rightarrow \dfrac{2}{2y} = \dfrac{a+b}{a-b}$

$\Rightarrow \dfrac{1}{y} = \dfrac{a+b}{a-b}$

$\Rightarrow \; y = \dfrac{a-b}{a+b}$ ...(ii)

Subtracting equation (ii) from (i) we get

$\therefore \; x - y = \dfrac{a+b}{a-b} - \dfrac{a-b}{a+b}$

$\left\{\because (a+b)^2 - (a-b)^2 = 4ab \text{ and } a^2 - b^2 = (a-b)(a+b)\right\}$

$= \dfrac{(a+b)^2 - (a-b)^2}{(a+b)(a-b)} = \dfrac{4ab}{a^2 - b^2}$

Multiply equation (i) and (ii) we get

$xy = \dfrac{a+b}{a-b} \times \dfrac{a-b}{a+b} = 1$

$\therefore$ Expression

$= \dfrac{x-y}{1+xy} = \dfrac{\frac{4ab}{a^2-b^2}}{1+1}$

$= \dfrac{4ab}{2\left(a^2 - b^2\right)} = \dfrac{2ab}{a^2 - b^2}$

525. Given $200 \div 25 \times 4 + 12 - 3$

$= \dfrac{200}{25} \times 4 + 12 - 3$

$= 8 \times 4 + 12 - 3$

$= 32 + 12 - 3 = 44 - 3 = 41$

526. Let the number be a.

Given $14 \times 627 \div \sqrt{1089} = (a)^3 + 141$

$14 \times 627 \div \sqrt{1089} = a^3 + 141$

$\Rightarrow 14 \times 627 \div 33 = a^3 + 141$

$\Rightarrow 14 \times 19 = a^3 + 141$

$\Rightarrow 266 = a^3 + 141$

$\Rightarrow a^3 = 125$

$\Rightarrow a = \sqrt[3]{125}$

$\Rightarrow a = 5$

527. Let the number be x.

Given $(923 - 347)/x = 32$

$\Rightarrow x = \dfrac{576}{32} = 18$

$\Rightarrow x = 18$

528. Let the number be a.

Given $1559.95 - 7.99 \times 24.96 - a^2 = 1154$

$\Rightarrow 1560 - 8 \times 25 - a^2 = 1154$

$\Rightarrow 1360 - 200 - a^2 = 1154$